



OFFLINE-FIRST SECURITY ENGINEERING

VNAGY Security Architecture and Behavioral Detection Standard

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1. Introduction & Scope

This document defines the **VNAGY Security Architecture and Behavioral Detection Standard**, a unified technical specification that consolidates the core principles, policies, heuristic models, and operational directives of the VNAGY system. The standard provides a complete description of VNAGY's offline-first security model, behavior-driven detection logic, and modular architectural design intended for deterministic, local, and reproducible signal analysis.

The scope of this standard includes six foundational components:

- **AI Isolation Policy (AIP-002)** — establishes strict boundaries preventing AI systems from influencing security decisions.
- **Modular Interaction Policy (MIP-003)** — defines deterministic communication rules between VNAGY modules.
- **Behavior Engine Constraints (BEC-005)** — specifies signal sources, baseline logic, drift detection, and behavior-based limitations.
- **Heuristic Scoring Model (HSM-004)** — describes the physics-inspired heuristics, scoring formulas, and multi-signal correlation rules.
- **Killing Chain Format (KCF-006)** — standardizes attack phases, KEY tool mapping, and MITRE-aligned scoring.
- **Offline-First Directive (OFD-001)** — defines VNAGY's operational model based entirely on local processing, zero cloud dependency, and full autonomy.

The document covers:

- system architecture and module boundaries
- security policies and operational directives
- heuristic and behavioral detection models
- scoring logic and correlation mechanisms
- offline-first processing principles
- cross-sector use-case scenarios (cloud, finance, healthcare, logistics, public sector)

The document does **not** include:

- internal thresholds, weights, or anti-evasion logic
- implementation details that could compromise system integrity
- cloud-based scoring, telemetry, or remote dependencies
- institutional certifications or external compliance claims

The purpose of this standard is to provide a **coherent, technically rigorous, and research-grade specification** of the VNAGY system for security architects, researchers, and R&D teams seeking to understand or evaluate behavior-security methodologies and offline-first detection frameworks.

VNAGY

2. Core Architectural Principles

The **VNAGY** system is built on a set of foundational architectural principles that ensure deterministic behavior, offline autonomy, and strict modular separation. These principles define how the system processes signals, enforces security boundaries, and maintains operational integrity without reliance on external infrastructure.

2.1 Offline-First Operation

VNAGY operates entirely in an offline-first mode, meaning all detection, scoring, correlation, and decision-making occur locally. No cloud services, remote scoring pipelines, telemetry feeds, or external dependencies are used. This principle ensures:

- full autonomy of the detection engine
- reproducible and deterministic results
- immunity to cloud outages or remote manipulation
- strict control over data locality and privacy

The **Offline-First Directive (OFD-001)** formalizes this requirement and prohibits any cloud-based logic within the core engine.

2.2 Deterministic Modular Architecture

The system is composed of independent modules (L-Pack, C-Pack, P-Pack, S-Pack, M-Module, X-Pack), each responsible for a specific operational domain. Modules communicate through a controlled, deterministic interaction model defined in the **Modular Interaction Policy (MIP-003)**. Key architectural guarantees include:

- strict module boundaries
- predictable data flow
- no hidden or implicit interactions
- reproducible behavior across environments

This modularity enables isolated development, secure upgrades, and clear auditing of system behavior.

2.3 Behavior-Driven Detection

VNAGY relies on behavior-based detection rather than signatures, cloud reputation, or machine-learning black boxes. The **Behavior Engine Constraints (BEC-005)** define the five signal sources and the rules governing baseline modeling, drift detection, and anomaly scoring. This approach ensures:

- detection of novel and unknown threats
- resistance to evasion techniques
- consistent scoring independent of external datasets
- physics-inspired heuristics that reflect real system behavior

Behavior-driven detection is the core of **VNAGY**'s security philosophy.

2.4 Physics-Inspired Heuristic Scoring

The **Heuristic Scoring Model (HSM-004)** defines how signals are evaluated using deterministic, physics-inspired heuristics. These heuristics model system behavior through concepts such as drift, deviation, energy, and stability. The scoring model provides:

- mathematically grounded detection logic
- multi-signal correlation
- reproducible scoring outcomes
- transparent and auditable decision paths

No machine learning, probabilistic inference, or opaque models are used.

2.5 Killing Chain Alignment

The **Killing Chain Format (KCF-006)** standardizes how VNAGY maps detected behaviors to attack phases. Each phase is associated with specific KEY tools and MITRE-aligned descriptions. This ensures:

- consistent interpretation of detected events
- structured escalation logic
- clear mapping between behavior and threat phase
- compatibility with industry frameworks

The Killing Chain provides the backbone for VNAGY's threat classification.

2.6 AI Isolation

The **AI Isolation Policy (AIP-002)** enforces strict separation between AI components and security-critical logic. AI may assist with documentation, summarization, or non-critical tasks, but it cannot:

- influence scoring
- modify heuristics
- alter thresholds
- participate in detection logic

This guarantees that VNAGY remains deterministic, auditable, and free from AI-driven unpredictability.

3. System Architecture Overview

The **VNAGY** system is designed as a fully offline-first, modular security framework composed of deterministic components that operate autonomously and without external dependencies. Its architecture is structured around clearly defined module boundaries, a controlled interaction model, and a behavior-driven detection pipeline. This section provides an overview of the system's core architectural elements and their operational relationships.

3.1 Modular System Structure

VNAGY is organized into six primary module groups, each responsible for a distinct operational domain. These modules form a layered architecture that ensures isolation, reproducibility, and predictable system behavior.

- **L-Pack** — Local system introspection and low-level signal acquisition.
- **C-Pack** — Core processing utilities and deterministic transformation logic.
- **P-Pack** — Process analysis, memory-only detection, and behavioral extraction.
- **S-Pack** — Signal normalization, baseline modeling, and drift evaluation.
- **M-Module** — Central behavior engine responsible for scoring, correlation, and Killing Chain mapping.
- **X-Pack** — User-facing interface layer (CLI/UI) that interacts with lower modules without influencing detection logic.

Each module operates independently and communicates through deterministic channels defined in the **Modular Interaction Policy (MIP-003)**.

3.2 Controlled Interaction Model

The interaction model ensures that modules exchange data in a predictable and auditable manner. All communication follows strict rules:

- no implicit or hidden data flow
- no cross-module mutation of internal state
- no AI-driven or probabilistic decision paths
- no cloud-based or remote dependencies

This model guarantees that VNAGY behaves identically across environments, enabling reproducible research and deterministic detection outcomes.

3.3 Local Event Bus

At the core of the architecture is the **local event bus**, a lightweight communication backbone that transports normalized signals between modules. The event bus:

- operates entirely offline
- enforces strict message formats
- prevents unauthorized module interaction
- ensures consistent timing and ordering of events

It is the primary mechanism through which signals reach the **M-Module** for scoring and correlation.

3.4 Behavior Engine Pipeline

The behavior engine pipeline is the central processing flow that transforms raw system signals into structured security decisions. It consists of:

1. **Signal Acquisition** — via L-Pack and P-Pack
2. **Normalization & Baseline Modeling** — via S-Pack
3. **Heuristic Evaluation** — via the **Heuristic Scoring Model (HSM-004)**
4. **Correlation & Drift Analysis** — via M-Module
5. **Killing Chain Mapping** — via the **Killing Chain Format (KCF-006)**
6. **Final Output** — delivered to X-Pack for user visibility

This pipeline is strictly deterministic and governed by the **Behavior Engine Constraints (BEC-005)**.

3.5 Offline-First Processing

All architectural components operate under the **Offline-First Directive (OFD-001)**, which mandates:

- zero cloud dependency
- zero remote scoring
- zero telemetry ingestion
- full local autonomy

This ensures that **VNAGY** remains secure, reproducible, and resistant to external manipulation.

3.6 Security Boundary Enforcement

Security boundaries are enforced through:

- strict module isolation
- deterministic communication rules
- immutable scoring logic
- AI isolation via the **AI Isolation Policy (AIP-002)**
- controlled access to internal state
- reproducible behavior across executions

These boundaries guarantee that **VNAGY**'s detection logic cannot be influenced by external systems, AI components, or cloud-based services.

3.7 Cross-Sector Use-Case Integration

The architecture supports multiple operational domains, including:

- cloud environments
- finance
- healthcare
- logistics
- public sector

Each domain is represented through structured use-case scenarios that demonstrate how VNAGY modules operate under different threat conditions. These scenarios do not modify the core architecture; they simply illustrate its application.

4. Security Policies

This section consolidates all **VNAGY** security policies that define system boundaries, module interaction rules, AI isolation requirements, behavior-engine constraints, heuristic scoring principles, and the standardized Killing Chain format. Each policy is included in its original form to preserve technical accuracy and ensure consistent interpretation across research, engineering, and operational environments.

4.1. AI Isolation Policy (AIP-002)

4.1.1. Policy Identifier

VNAGY-AIP-002 — AI Isolation Policy

4.1.2. Directive Statement

This policy defines mandatory isolation rules for AI-adjacent components within the **VNAGY** framework.

4.1.3. Mandatory Rules

- Components **MUST** operate under strict isolation boundaries.
- AI-adjacent modules **MUST** maintain deterministic state transitions.
- Internal symbolic states **MUST** remain non-exposed.
- Local scoring **MUST** remain offline-valid.

4.1.4. Prohibited Actions

- Components **MUST NOT** access external networks.
- Modules **MUST NOT** share internal operational state.
- AI-adjacent logic **MUST NOT** initiate cross-module communication.
- No component **MUST NOT** expose thresholds, algorithms, or dynamic logic.

4.1.5. Allowed Actions

- Symbolic scoring **MAY** be performed locally.
- Deterministic transitions **MAY** be logged.
- Static configuration **MAY** be referenced.

4.1.6. Interaction Boundaries

- Modules **MAY** interact only through predefined symbolic channels.
- **No direct operational linkage between AI-adjacent components is permitted.**

4.1.7. Enforcement Notes

- Violations invalidate module compliance.
- Non-compliant modules **SHALL** be excluded from **VNAGY** integration.

4.1.8. Non-Operational Clause

This policy defines no operational logic, thresholds, algorithms, or reconstructable behavior.

4.2. Modular Interaction Policy (MIP-003)

4.2.1. Policy Identifier

VNAGY-MIP-003 — Modular Interaction Policy

4.2.2. Directive Statement

This policy defines mandatory interaction boundaries for modules within the **VNAGY** framework.

4.2.3. Mandatory Rules

- Modules **MUST** interact exclusively through predefined symbolic channels.
- Module state transitions **MUST** remain deterministic and isolated.
- Internal symbolic state **MUST** remain non-exposed to external modules.
- Module interfaces **MUST** remain static and non-dynamic.
- All interactions **MUST** remain offline-valid.

4.2.4. Prohibited Actions

- Modules **MUST NOT** access or modify another module's internal state.
- Modules **MUST NOT** initiate direct operational communication.
- No module **MUST NOT** expose thresholds, algorithms, or dynamic logic.
- Modules **MUST NOT** rely on network availability for interaction.
- **Cross-module operational dependencies MUST NOT exist.**

4.2.5. Allowed Actions

- Modules **MAY** exchange symbolic tokens through approved channels.
- Modules **MAY** reference static configuration data.
- **Deterministic symbolic scoring MAY be performed locally.**

4.2.6. Interaction Boundaries

- All module interactions **SHALL** occur through predefined symbolic interfaces.
- No operational linkage between modules is permitted.
- Interaction channels **SHALL** remain non-reconstructable and non-algorithmic.

4.2.7. Enforcement Notes

- Violations invalidate module compliance.
- Non-compliant modules **SHALL** be excluded from **VNAGY** integration.

4.2.8. Non-Operational Clause

This policy defines no operational logic, thresholds, algorithms, or reconstructable behavior.

4.3. Heuristic Scoring Model (HSM-004)

4.3.1. Policy Identifier

VNAGY-HSM-004 — Heuristic Scoring Model

4.3.2. Constraint Statement

This document defines the conceptual structure, symbolic boundaries, and deterministic constraints of the **VNAGY** Heuristic Scoring Model (HSM). It does **not** define operational logic, thresholds, algorithms, or reconstructable scoring behavior.

4.3.3. Mandatory Rules

- The model **MUST** operate exclusively on symbolic, non-numeric, non-algorithmic representations.
- All scoring outcomes **MUST** remain deterministic under identical symbolic conditions.
- Internal model states **MUST** remain isolated and non-exposed.
- All transformations **MUST** remain offline-valid and locally resolvable.

4.3.4. Prohibited Actions

- The model **MUST NOT** define numeric thresholds, weights, or statistical operations.
- The model **MUST NOT** expose internal scoring logic, decision paths, or reconstructable behavior.
- The model **MUST NOT** perform machine-learning, pattern-matching, or predictive operations.
- The model **MUST NOT** depend on runtime variability or external entropy sources.
- The model **MUST NOT** output operational decisions.
- The model **MUST NOT** rely on external systems, networks, or dynamic data sources.

4.3.5. Allowed Actions

- The model **MAY** use symbolic tokens to represent conceptual states.
- The model **MAY** define deterministic symbolic transitions.
- The model **MAY** produce non-operational, non-actionable symbolic outcomes.

4.3.6. Interaction Boundaries

- All scoring **SHALL** remain symbolic, deterministic, and non-numeric.
- No reconstructable logic **SHALL** be present.
- No operational linkage to other modules **SHALL** exist.
- Model outputs **SHALL** remain advisory, non-actionable, and non-binding.

4.3.7. Enforcement Notes

- Any violation of symbolic, deterministic, or non-operational constraints invalidates HSM compliance.
- Non-compliant implementations **SHALL** be excluded from **VNAGY** integration.

4.3.8. Non-Operational Clause

This specification defines no algorithms, no thresholds, no numeric scoring, and no reconstructable decision logic. It is strictly conceptual and normative.

4.4. Killing Chain Format

4.4.1. Policy Identifier

VNAGY-KCF-006 — Killing Chain Format

4.4.2. Format Statement

This document defines the conceptual structure, symbolic representation rules, and deterministic constraints of the **VNAGY** Killing Chain Format (KCF). It does **not** define operational logic, detection algorithms, thresholds, or reconstructable incident workflows.

4.4.3. Mandatory Rules

- The format **MUST** represent events and states as symbolic, non-numeric, non-algorithmic tokens.
- All chain transitions **MUST** remain deterministic under identical symbolic input.
- Internal chain structure and ordering **MUST** remain non-exposed and non-derivable from outputs.
- All representations **MUST** remain offline-valid and locally resolvable

4.4.4. Prohibited Actions

- The format **MUST NOT** define numeric severity, probability, or risk scores.
- The format **MUST NOT** encode detection logic, correlation rules, or operational playbooks.
- The format **MUST NOT** expose internal ordering logic, decision paths, or reconstructable workflows.
- The format **MUST NOT** perform aggregation, prediction, or pattern recognition.
- The format **MUST NOT** produce actionable or operational instructions.
- The format **MUST NOT** depend on external systems, dynamic data, or runtime variability.

4.4.5. Allowed Actions

- The format **MAY** define symbolic stages representing conceptual phases of an attack chain.
- The format **MAY** define deterministic symbolic transitions between stages.
- The format **MAY** represent non-operational, advisory, symbolic incident descriptors.

4.4.6. Interaction Boundaries

- All content **SHALL** remain symbolic, deterministic, and non-numeric.
- No reconstructable detection or response logic **SHALL** be present.
- No operational linkage to monitoring, response, or enforcement systems **SHALL** exist.
- Outputs **SHALL** remain advisory, non-actionable, and non-binding.

4.4.7. Enforcement Notes

- Any inclusion of operational logic, thresholds, or reconstructable workflows invalidates KCF compliance.
- Non-compliant formats **SHALL** be excluded from **VNAGY** integration.

4.4.8. Non-Operational Clause

This specification defines no detection algorithms, no response procedures, no numeric scoring, and no reconstructable incident handling logic. It is strictly conceptual and normative.

4.6 Offline-First Directive (OFD-001)

4.6.1. Policy Identifier

VNAGY-OFD-001 — Offline-First Directive

4.6.2. Directive Statement

This directive defines mandatory offline-first operational constraints for all **VNAGY** components.

4.6.3. Mandatory Rules

- Components **MUST** operate correctly without network access.
- All logic **MUST** remain fully deterministic under offline conditions.
- State transitions **MUST NOT** depend on remote systems.
- Local symbolic scoring **MUST** remain valid without external resources.
- Configuration **MUST** remain static and locally resolvable.

4.6.4. Prohibited Actions

- Components **MUST NOT** initiate or expect remote communication.
- Modules **MUST NOT** rely on cloud services, APIs, or remote storage.
- No component **MUST NOT** degrade functionality when offline.
- Operational logic **MUST NOT** include dynamic network-dependent behavior.
- Components **MUST NOT** expose network-based thresholds or algorithms.

4.6.5. Allowed Actions

- Components **MAY** reference static local configuration files.
- Deterministic symbolic scoring **MAY** be performed locally.
- Modules **MAY** log offline-valid symbolic events.

4.6.6. Interaction Boundaries

- All interactions **SHALL** remain local and symbolic.
- No remote operational linkage is permitted.
- Offline-first behavior **SHALL** remain non-reconstructable and non-algorithmic.

4.6.7. Enforcement Notes

- Violations invalidate offline-first compliance.
- Non-compliant components **SHALL** be excluded from **VNAGY** integration.

4.6.8. Non-Operational Clause

This policy defines no operational logic, thresholds, algorithms, or reconstructable behavior.

5. Behavior Engine Specification

The **VNAGY** Behavior Engine is the core analytical component responsible for transforming raw system signals into structured, deterministic security decisions. It operates entirely offline, adheres to strict module boundaries, and evaluates system behavior using physics-inspired heuristics defined in the **Heuristic Scoring Model (HSM-004)**. This section provides a detailed specification of the engine's operational flow, constraints, and evaluation logic.

5.1. Engine Purpose and Design Principles

The Behavior Engine is designed to:

- evaluate system behavior deterministically
- detect anomalies through drift, deviation, and instability
- correlate multi-source signals into unified security outcomes
- map detected behaviors to attack phases using the **Killing Chain Format (KCF-006)**
- operate fully offline under the **Offline-First Directive (OFD-001)**

The engine does **not** rely on machine learning, cloud telemetry, probabilistic

inference, or AI-driven decision paths.

5.2. Signal Sources

The engine processes five deterministic signal sources defined in the **Behavior Engine Constraints (BEC-005)**:

- **CPU Behavior** — load stability, unexpected spikes, anomalous scheduling.
- **Memory Behavior** — memory-only processes, injection patterns, allocation anomalies.
- **Process Behavior** — execution flow, parent-child anomalies, stealth processes.
- **Network Behavior** — outbound patterns, subnet deviation, unexpected persistence.
- **System Events** — file operations, privilege changes, configuration drift.

Each signal is normalized by S-Pack before entering the engine.

5.3. Baseline Modeling

The engine maintains deterministic baselines for:

- CPU stability
- memory allocation patterns
- process lifecycle behavior
- network subnet heuristics
- system event frequency

Baselines are updated locally and never depend on cloud datasets or external reputation systems.

Baseline deviation is evaluated using physics-inspired heuristics such as:

- drift
- energy
- momentum
- stability

- deviation amplitude

These concepts are defined in the **Heuristic Scoring Model (HSM-004)**.

5.4. Heuristic Evaluation

Each signal is evaluated using deterministic heuristics that quantify:

- deviation from baseline
- rate of change
- correlation with other signals
- temporal consistency
- anomaly persistence

The heuristics produce intermediate scores that are then combined into a unified behavior score.

No thresholds, weights, or anti-evasion logic are included in this document, in accordance with VNAGY security requirements.

5.5. Multi-Signal Correlation

The Behavior Engine correlates signals across all five domains to detect:

- coordinated anomalies
- multi-phase attack behavior
- stealthy or memory-only execution patterns
- cross-domain drift
- persistent deviation signatures

Correlation rules are deterministic and defined in the **Behavior Engine Constraints (BEC-005)**.

5.6. Killing Chain Mapping

Once correlation is complete, the engine maps detected behavior to the appropriate attack phase using the **Killing Chain Format (KCF-006)**. This mapping:

- standardizes threat interpretation
- aligns behavior with MITRE-style descriptions

- identifies KEY tools associated with each phase
- provides structured escalation logic

The Killing Chain does not modify scoring; it interprets it.

5.7. Output Generation

The final output of the Behavior Engine includes:

- unified behavior score
- correlated signal summary
- Killing Chain phase classification
- module-safe metadata for X-Pack display

Outputs are deterministic, reproducible, and fully offline.

5.8. AI Isolation

The engine operates under strict AI isolation rules defined in the **AI Isolation Policy (AIP-002)**. AI components may:

- summarize results
- assist with documentation
- provide non-critical insights

AI components may **not**:

- influence scoring
- modify heuristics
- alter baselines
- participate in detection logic

This ensures full determinism and auditability.

6. Heuristic Scoring Model

The Heuristic Scoring Model (HSM-004) defines the deterministic, physics-inspired evaluation logic used by the VNAGY Behavior Engine to quantify system behavior. Unlike probabilistic or machine-learning approaches, VNAGY relies exclusively on transparent, reproducible heuristics that measure deviation, drift, stability, and correlation across multiple signal domains. This section outlines the conceptual framework, evaluation rules, and scoring flow that govern heuristic-based detection.

6.1 Model Purpose and Philosophy

The heuristic scoring model is designed to:

- quantify behavioral deviation using deterministic rules
- evaluate system stability through physics-inspired concepts
- correlate multi-domain signals into unified scores
- ensure reproducible and auditable detection outcomes
- operate fully offline under the **Offline-First Directive (OFD-001)**

The model does **not** use machine learning, cloud datasets, or probabilistic inference.

6.2 Physics-Inspired Concepts

The scoring model uses physics-inspired abstractions to evaluate system behavior. These concepts provide a structured way to measure deviation and drift:

- **Drift** — gradual deviation from baseline over time
- **Energy** — intensity of behavioral change
- **Momentum** — persistence of deviation across intervals
- **Stability** — resistance to anomalous fluctuation
- **Amplitude** — magnitude of deviation from expected behavior

These concepts are deterministic and defined in the **Heuristic Scoring Model (HSM-004)**.

6.3 Input Signals

The scoring model evaluates normalized signals from five domains:

- **CPU Behavior**
- **Memory Behavior**
- **Process Behavior**
- **Network Behavior**
- **System Events**

All signals are pre-processed by S-Pack to ensure consistent formatting and baseline alignment.

6.4 Baseline Deviation Measurement

Each signal is compared against its deterministic baseline. Deviation is measured using:

- rate of change
- deviation amplitude
- temporal persistence
- cross-domain correlation
- stability loss

Baselines are updated locally and never depend on cloud telemetry or external reputation systems.

6.5 Heuristic Evaluation Rules

The scoring model applies a set of deterministic rules to evaluate each signal:

- **Rule 1: Drift Detection** Identify gradual deviation from baseline across time windows.
- **Rule 2: Instability Detection** Detect sudden fluctuations inconsistent with normal system behavior.
- **Rule 3: Persistence Analysis** Evaluate whether anomalies persist across multiple intervals.
- **Rule 4: Cross-Domain Correlation** Combine signals from different domains to detect coordinated behavior.
- **Rule 5: Temporal Consistency** Assess whether anomalies align with known attack timing patterns.

These rules produce intermediate scores for each signal domain.

6.6 Multi-Signal Scoring

Intermediate scores are combined into a unified behavior score using deterministic correlation logic defined in the **Behavior Engine Constraints (BEC-005)**. The correlation process evaluates:

- alignment of anomalies across domains
- synchronized deviation patterns
- multi-phase behavioral signatures
- memory-only or stealth execution indicators

The unified score represents the overall behavioral anomaly level.

6.7 Scoring Output

The final scoring output includes:

- unified behavior score
- domain-specific deviation summaries
- drift and stability indicators
- correlation metadata
- escalation flags for Killing Chain mapping

This output is passed to the **M-Module** for final interpretation and phase classification.

6.8 Determinism and Reproducibility

The scoring model guarantees:

- identical results across repeated executions
- no influence from AI components
- no cloud-based variability
- full auditability of scoring logic

AI isolation is enforced by the **AI Isolation Policy (AIP-002)**.

7. Killing Chain Format

The Killing Chain Format (KCF-006) defines **VNAGY**'s standardized method for interpreting behavioral anomalies as structured attack phases. It provides a deterministic mapping between observed system behavior, heuristic scoring outputs, and the operational stages of an intrusion. Unlike probabilistic threat models, the Killing Chain is fully deterministic, reproducible, and aligned with VNAGY's offline-first and behavior-driven design principles.

The Killing Chain does **not** modify heuristic scores; it contextualizes them. It transforms raw behavioral deviation into a clear, phase-based representation of adversarial activity.

7.1 Purpose of the Killing Chain

The Killing Chain Format is designed to:

- provide a structured interpretation of detected behavior
- align anomalies with known attack phases
- support deterministic escalation logic
- ensure consistent threat classification across environments
- map behavior to KEY tools and MITRE-style descriptions
- maintain full offline autonomy and reproducibility

It acts as the interpretive layer between the **Behavior Engine** and the user-facing interface.

7.2 Killing Chain Phases

The **VNAGY** Killing Chain consists of deterministic phases that represent the progression of adversarial activity. Each phase is associated with specific behavioral signatures and KEY tools.

Phase 1 — Reconnaissance

Behavioral indicators include:

- unusual enumeration patterns
- abnormal system queries
- unexpected process introspection Mapped via: **CPU Behavior** and **System Events**.

Phase 2 — Initial Access

Indicators include:

- anomalous process creation
- unexpected privilege transitions
- memory-only execution attempts Mapped via: **Process Behavior** and **Memory Behavior**.

Phase 3 — Execution

Indicators include:

- sustained deviation in execution flow
- drift in CPU scheduling
- stealth or injected processes Mapped via: **CPU Behavior** and **Process Behavior**.

Phase 4 — Persistence

Indicators include:

- repeated anomalous patterns
- stable deviation signatures
- long-term drift Mapped via: **Baseline Modeling** and **Drift Analysis**.

Phase 5 — Command & Control

Indicators include:

- outbound network anomalies
- subnet deviation
- unexpected persistence in communication Mapped via: **Network Behavior**.

Phase 6 — Impact

Indicators include:

- destructive or destabilizing behavior
- high-amplitude deviation
- correlated multi-domain anomalies Mapped via: **Unified Behavior Score** and **Correlation Logic**.

7.3 KEY Tools Mapping

Each phase is associated with specific KEY tools that represent the operational capabilities of an adversary. KEY tools are not executable components; they are conceptual markers used to classify behavior.

Examples include:

- **KEY-ENUM** — enumeration behavior
- **KEY-EXEC** — execution behavior
- **KEY-MEM** — memory-only behavior
- **KEY-NET** — network deviation
- **KEY-PERSIST** — persistence behavior

KEY tools provide a structured vocabulary for interpreting behavior without exposing internal thresholds or anti-evasion logic.

7.4 Deterministic Phase Mapping

The Killing Chain maps behavior to phases using deterministic rules:

- **Rule 1:** A phase is assigned only when deviation is confirmed across its relevant domain.
- **Rule 2:** Phases escalate only when anomalies persist or correlate across domains.
- **Rule 3:** Phase transitions do not depend on external datasets or cloud telemetry.
- **Rule 4:** AI components cannot influence phase classification (per **AIP-002**).
- **Rule 5:** Mapping is reproducible across all environments.

These rules ensure that **VNAGY**'s threat interpretation is stable, predictable, and auditable.

7.5 Integration with the Behavior Engine

The Killing Chain receives:

- unified behavior score
- domain-specific deviation summaries
- correlation metadata
- drift and stability indicators

It then produces:

- phase classification
- KEY tool mapping
- structured escalation logic
- final threat interpretation for X-Pack display

The Killing Chain does **not** modify scores; it interprets them.

7.6 Offline-First Interpretation

All phase mapping occurs locally, without:

- cloud scoring
- remote telemetry
- external datasets
- AI-driven inference

This ensures that threat interpretation remains deterministic and secure under the **Offline-First Directive (OFD-001)**.

8. Use-Case Scenarios

This section presents domain-specific use-case scenarios that demonstrate how the VNAGY system operates under different threat conditions. Each scenario illustrates the application of VNAGY's offline-first architecture, behavior-driven detection, heuristic scoring, and Killing Chain mapping within a particular industry context. These scenarios do not modify the core system design; they serve as practical examples of how VNAGY interprets and responds to real-world behavioral anomalies.

8.1. Cloud Environment Scenario

Cloud environments exhibit dynamic workloads, ephemeral processes, and frequent configuration changes. VNAGY's offline-first model ensures that detection remains autonomous even in distributed or containerized deployments.

Key behavioral indicators include:

- drift in container lifecycle behavior
- anomalous API invocation patterns
- unexpected privilege escalation within orchestration layers
- memory-only execution inside short-lived workloads

VNAGY correlates these signals using the **Behavior Engine** and maps them to phases such as **Reconnaissance**, **Execution**, or **Persistence** via the **Killing Chain Format**.

8.2. Finance Sector Scenario

Financial systems require strict stability, predictable process behavior, and strong resistance to unauthorized access. **VNAGY**'s physics-inspired heuristics detect subtle deviations that may indicate fraud, unauthorized automation, or stealthy intrusion.

Key behavioral indicators include:

- abnormal transaction-processing timing
- drift in CPU scheduling during peak hours
- memory-only automation scripts
- anomalous outbound communication to non-financial subnets

The **Heuristic Scoring Model** quantifies deviation, while the **Offline-First Directive** ensures no sensitive data leaves the environment.

8.3. Healthcare Sector Scenario

Healthcare systems often involve regulated environments, legacy devices, and critical uptime requirements. VNAGY's deterministic detection model is well-suited for environments where false positives or cloud dependency are unacceptable.

Key behavioral indicators include:

- unexpected access pattern changes in medical systems
- drift in device communication frequency
- anomalous process behavior on diagnostic machines
- memory-only execution on systems with restricted storage

VNAGY maps these anomalies to phases such as **Initial Access** or **Command & Control**, depending on correlation strength.

8.4. Logistics Sector Scenario

Logistics environments rely on predictable workflows, route integrity, and stable device communication. **VNAGY** detects deviations that may indicate tampering, unauthorized automation, or compromised operational systems.

Key behavioral indicators include:

- drift in route-planning processes
- anomalous API usage in fleet management systems
- unexpected persistence of background tasks
- network deviation across operational subnets

The **Behavior Engine Constraints** ensure consistent detection even in distributed or low-connectivity environments.

8.5. Public Sector Scenario

Public sector systems often operate under strict regulatory requirements and require high transparency, auditability, and deterministic behavior. VNAGY's offline-first model ensures full autonomy and reproducibility.

Key behavioral indicators include:

- anomalous access to administrative interfaces
- drift in configuration management behavior
- unexpected privilege transitions
- coordinated multi-domain anomalies across public infrastructure systems

The **AI Isolation Policy** ensures that all detection remains deterministic and auditable.

8.6. Purpose of Use-Case Scenarios

These scenarios serve to:

- illustrate **VNAGY**'s cross-sector applicability
- demonstrate behavior-driven detection in real environments
- show how heuristic scoring interprets domain-specific anomalies
- provide structured examples for research, documentation, and evaluation
- maintain consistency with **VNAGY**'s offline-first and modular design principles

Use-case scenarios do not introduce new logic; they showcase how existing VNAGY components behave under different operational conditions.

9. Compliance & Future Work

This section outlines **VNAGY**'s alignment with industry security principles and presents the planned future enhancements that will expand the system's capabilities. **VNAGY** is designed as a deterministic, offline-first security architecture that adheres to transparent, auditable, and reproducible operational standards. While **VNAGY** does not claim certification from external institutions, its design philosophy aligns with widely recognized security frameworks and best practices.

9.1. Compliance Alignment

VNAGY's architecture and policies are conceptually aligned with several industry-standard security principles, including:

- **Deterministic Detection Models** — consistent with reproducibility requirements found in research-grade frameworks.
- **Offline-First Processing** — aligned with privacy-preserving and data-locality principles used in regulated environments.
- **Modular Isolation** — similar to architectural separation recommended in secure system design guidelines.
- **Behavior-Driven Detection** — compatible with modern approaches to anomaly detection and threat modeling.

- **Transparent Heuristics** — aligned with auditability requirements in high-assurance systems.

VNAGY does **not** rely on cloud telemetry, external scoring pipelines, or AI-driven inference, ensuring compliance with environments that require strict autonomy and data locality.

9.2. Internal Policy Compliance

VNAGY adheres to its own internal policies, which define the system's operational boundaries:

- **AI Isolation Policy (AIP-002)**
- **Modular Interaction Policy (MIP-003)**
- **Behavior Engine Constraints (BEC-005)**
- **Heuristic Scoring Model (HSM-004)**
- **Killing Chain Format (KCF-006)**
- **Offline-First Directive (OFD-001)**

These policies ensure that VNAGY remains deterministic, auditable, and resistant to external influence.

9.3. Future Work

VNAGY's development roadmap includes several planned enhancements that will expand the system's analytical capabilities while preserving its offline-first and deterministic nature.

9.3.1. Extended Signal Domains

Future versions may introduce additional behavior domains, such as:

- device-specific operational heuristics
- hardware-level anomaly detection
- extended memory-only behavioral signatures

9.3.2. Enhanced Correlation Models

Planned improvements include:

- deeper multi-domain correlation logic
- extended temporal analysis
- refined drift-stability interaction models

These enhancements will remain deterministic and transparent.

9.3.3. Expanded Use-Case Libraries

Additional domain scenarios will be added to support:

- industrial control systems
- critical infrastructure
- high-assurance environments
- regulated public-sector deployments

9.3.4. Formal Specification Extensions

Future versions may include:

- extended module documentation
- formal mathematical notation for heuristics
- expanded Killing Chain phase definitions
- structured metadata formats for X-Pack output

9.3.5. Research Integration

VNAGY will continue to evolve through:

- offline-first research experiments
- behavior-security model refinement
- cross-domain anomaly studies
- deterministic detection methodology development

All future work will maintain strict compliance with VNAGY's core principles and internal policies.

9.4. Commitment to Determinism and Transparency

VNAGY's future development will continue to prioritize:

- deterministic behavior
- offline autonomy
- transparent heuristics
- strict module isolation
- reproducible detection outcomes
- AI isolation and auditability

These principles ensure that VNAGY remains a stable, predictable, and research-grade security architecture.

VNAGY

10. Conclusion

The **VNAGY** Security Architecture and Behavioral Detection Standard provides a unified, deterministic, and offline-first framework for analyzing system behavior across diverse operational environments. Through its modular design, physics-inspired heuristics, strict AI isolation, and reproducible detection logic, **VNAGY** establishes a transparent and auditable foundation for behavior-driven security systems.

The architecture integrates core components such as the **M-Module**, **Heuristic Scoring Model**, **Behavior Engine Constraints**, and the **Killing Chain Format** to deliver consistent and interpretable security outcomes. Its offline-first philosophy ensures full autonomy, data locality, and independence from cloud-based scoring or external telemetry, making **VNAGY** suitable for regulated, high-assurance, and privacy-sensitive environments.

The use-case scenarios demonstrate how **VNAGY's** principles apply across sectors such as cloud, finance, healthcare, logistics, and public infrastructure, without altering the underlying architecture. These examples highlight the adaptability of **VNAGY's** behavior-security approach while maintaining strict determinism and module isolation.

Looking forward, **VNAGY's** development will continue to expand through enhanced correlation models, extended signal domains, and formal specification improvements — all while preserving the system's core values of transparency, reproducibility, and offline autonomy. The standard serves as a foundation for future research, engineering work, and practical deployment of behavior-driven security systems.

VNAGY remains committed to delivering a stable, predictable, and technically rigorous security architecture built on clear principles, structured policies, and deterministic detection logic.

11. Intellectual Property & Licensing

11.1. Ownership

The **VNAGY Security Architecture and Behavioral Detection Standard**, including all symbolic constructs, interaction rules, structural definitions, and conceptual models, is the exclusive intellectual property of the author.

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11.4. Non-Operational Declaration

This document contains no operational details, no thresholds, no weights, no heuristics, no correlation logic, and no algorithmic semantics. All constructs are symbolic and conceptual. The model is non-computational and non-reconstructable by design.

11.5. License

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<https://github.com/VNAGY-sec-ViktorijaN/VNAGY-Framework/tree/main/VNAGY/LICENSE>

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